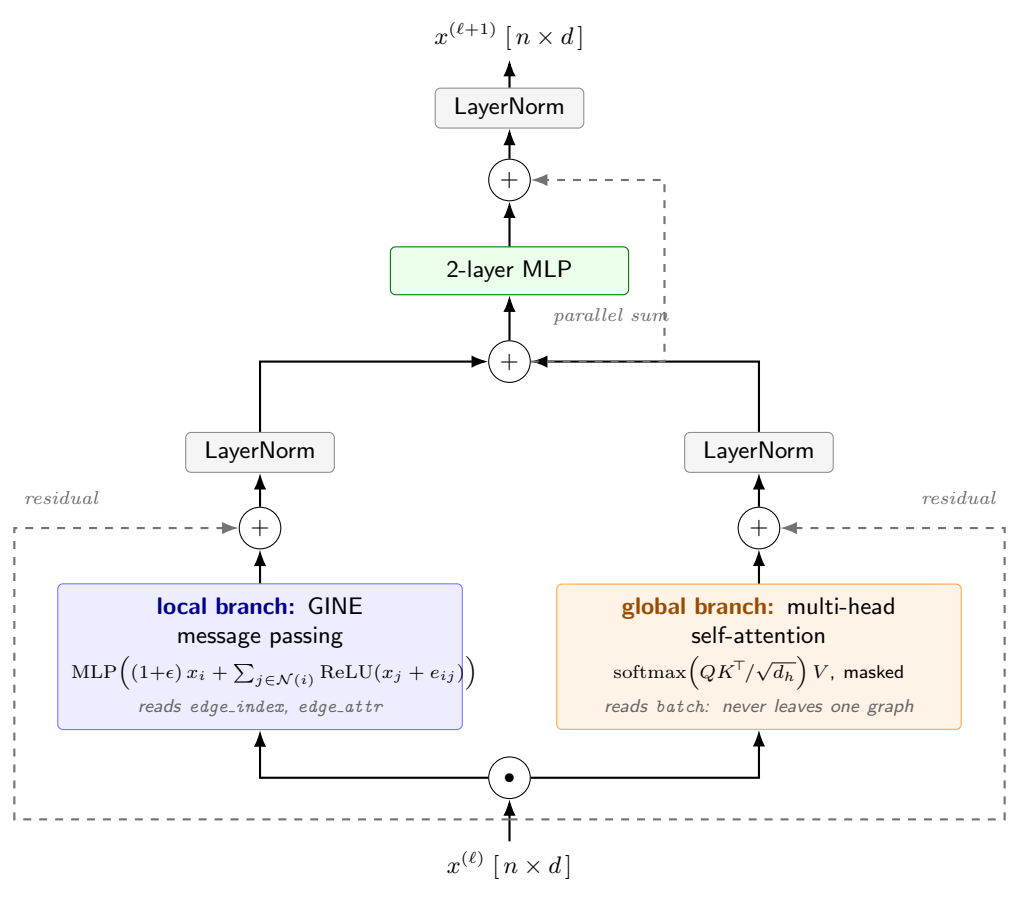

$x^{(\ell+1)}$ $[n \times d]$
LayerNorm
2-layer MLP
parallel sum
LayerNorm
LayerNorm
residual
residual
local branch: GINE
message passing
$\mathrm{MLP}\left((1+\epsilon)\, x_i + \sum_{j \in \mathcal{N}(i)} \mathrm{ReLU}(x_j + e_{ij})\right)$
reads edge_index, edge_attr
global branch: multi-head
self-attention
$\mathrm{softmax}\left(QK^\top/\sqrt{d_h}\right) V$, masked
reads batch: never leaves one graph
$x^{(\ell)}$ $[n \times d]$